

Project Coversheet

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Project Title	Week 3: Churn Prediction for StreamWorks Media

Instructions:

Students must download this cover sheet, use it as the first page of their project, and then save the entire document as a PDF before submission.

Project Guidelines and Rules

1. Formatting and Submission

- Format: Use a readable font (e.g., Arial/Times New Roman), size 12, 1.5 line spacing.
- Title: Include Week and Title (Example - Week 1: Travel Ease Case Study.)
- File Format: Submit as PDF or Word file
- Page Limit: 4–5 pages, including the title and references.

2. Answer Requirements

- Word Count: Each answer should be within 100–150 words; Maximum 800–1,200 words.
- Clarity: Write concise, structured answers with key points.
- Tone: Use formal, professional language.

3. Content Rules

- Answer all questions thoroughly, referencing case study concepts.
- Use examples where possible (e.g., risk assessment techniques).
- Break complex answers into bullet points or lists.

4. Plagiarism Policy

- Submit original work; no copy-pasting.
- Cite external material in a consistent format (e.g., APA, MLA).

5. Evaluation Criteria

- Understanding: Clear grasp of business analysis principles.
- Application: Effective use of concepts like cost-benefit analysis and Agile/Waterfall.
- Clarity: Logical, well-structured responses.
- Creativity: Innovative problem-solving and examples.
- Completeness: Answer all questions within the word limit.

6. Deadlines and Late Submissions

- Deadline: Submit on time; trainees who fail to submit the project will miss the “Certificate of Excellence”

7. Additional Resources

- Refer to lecture notes and recommended readings.
- Contact the instructor or peers for clarifications before the deadline.

YOU CAN START YOUR PROJECT FROM HERE

Introduction

StreamWorks Media is a fast-growing UK based video streaming platform operating in a highly competitive market alongside global providers such as Netflix and Amazon Prime. As customer acquisition costs rise and competition intensifies, retaining existing subscribers has become a critical priority for the business. Understanding why users cancel their subscriptions is essential to protecting revenue and improving long term customer value.

Our goal is to understand who is churning, why they churn, and how churn impacts revenue and engagement. In addition, the project seeks to predict churn probability so that the business can take early, targeted retention actions.

Data Cleaning Summary

The dataset underwent thorough cleaning to ensure accuracy, consistency, and suitability for churn analysis across all customer related fields. The dataset contains a total of 1,500 records.

RangeIndex: 1500 entries, 0 to 1499					
Data columns (total 14 columns):					
#	Column	Non-Null	Count	Dtype	
0	user_id	1498	non-null	float64	user_id
1	age	1497	non-null	float64	age
2	gender	1499	non-null	object	gender
3	signup_date	1498	non-null	object	signup_date
4	last_active_date	1498	non-null	object	last_active_date
5	country	1497	non-null	object	country
6	subscription_type	1497	non-null	object	subscription_type
7	average_watch_hours	1496	non-null	float64	average_watch_hours
8	mobile_app_usage_pct	1498	non-null	float64	mobile_app_usage_pct
9	complaints_raised	1497	non-null	float64	complaints_raised
10	received_promotions	1497	non-null	object	received_promotions
11	referred_by_friend	1497	non-null	object	referred_by_friend
12	is_churned	1499	non-null	float64	is_churned
13	monthly_fee	1355	non-null	float64	monthly_fee

Data Types

- Two fields, user_id and age, initially had incorrect float data types. After handling missing values, both fields were converted to integer.

Missing values

- Missing data was present across multiple columns, with monthly fee showing the highest number of missing values at **145 records**. Other missing values included age 3, gender 1, signup_date 2, last_active_date 2, country 3, subscription_type 3, average_watch_hours 4, mobile_app_usage_pct 2, complaints_raised 3, received_promotions 3, referred_by_friend 3, and is_churned 1.
- Age was imputed using the mode value. Gender, subscription_type, and country missing values were replaced with Not Selected to preserve records. Complaints_raised and monthly_fee were retained as missing due to business uncertainty around appropriate imputation. The single missing is_churned value was noted but overall target completeness remained strong.

Standardizations

- Categorical columns including gender, subscription_type, received_promotions, and referred_by_friend were standardized by fixing casing, trimming spaces, and correcting spelling inconsistencies. This ensured consistent category definitions across the dataset. Subscription types and loyalty related fields were label encoded due to their ordered nature, while gender, country, promotions, and referrals were one hot encoded as nominal variables.

Invalid entries

- Date fields signup_date and last_active_date were converted to datetime format. A total of 48 records were identified where last_active_date occurred before signup_date, resulting in negative tenure values. These negative tenure values were replaced with nulls to avoid misleading behavioral metrics.

Numeric validation

- Numeric columns such as age, average_watch_hours, mobile_app_usage_pct, complaints_raised, and monthly_fee were converted to numeric formats to enforce consistency. Invalid entries were coerced to NaN. Ranges were checked to ensure no negative or unrealistic values remained. The is_churned variable was converted to numeric and confirmed to have only one missing value, indicating good data quality for the target variable.

Overall Insight

The dataset is generally of good quality with minimal duplicates and mostly complete target variable values. Missing values exist across several columns, with `monthly_fee` showing the highest gaps, but key identifiers and target fields remain intact. Categorical data was standardized and encoded appropriately, supporting consistent analysis. Date and numeric fields were validated, with negative or invalid entries handled carefully to prevent misleading metrics. Overall, the dataset is reliable for analysis after the applied cleaning and standardization steps.

Feature Engineering Summary

Encoding

- Label encoding was applied to ordered categorical variables such as subscription type and loyalty tiers, converting each category into a unique numeric value. One hot encoding was used for nominal variables including gender, country, received promotions, and referred by friend, creating binary indicator columns to accurately represent categories with no inherent order.

New Features

- **Tenure Days**
 - The tenure of users in the dataset ranges widely, with a mean of 561 days and a standard deviation of 323 days, indicating moderate variability in user engagement. Most users (1446 out of 1500) have valid tenure values, with the middle 50 percent of users active between 283 and 821 days, and extreme values ranging from a minimum of 6 days to a maximum of 1281 days. A small subset of 48 records had negative tenure due to `last_active_date` occurring before `signup_date`, which were replaced with nulls to prevent misleading analysis, ensuring the dataset accurately reflects user activity patterns and supports reliable behavioral insights.
- **Loyalty Tiers**
 - Among customers with tenure above 180 days, loyalty tier data is available for 1498 out of 1500 users, showing high coverage. The dataset contains two unique values, True and False, indicating whether a customer exceeds 180 days of tenure, with True being the most common. A total of 1228 customers fall into the True category, demonstrating that the majority of longer-tenured users are considered loyal,

highlighting a strong correlation between tenure duration and loyalty within this customer base.

- **Watch per fee Ratio**

- The watch-per-fee ratio, calculated using monthly fee and average watch hours, shows significant variation across 1349 valid records. The ratio ranges from a minimum of 0.03 to a maximum of 13.34, with a mean of 4.45 and a standard deviation of 3.20, indicating diverse usage patterns relative to cost. The median ratio of 3.88 and interquartile range from 2.00 to 6.11 suggest that most users consume between two to six hours per unit of fee.

- **Heavy mobile users**

- Mobile app usage analysis shows that out of 1498 valid records, there are two categories indicating heavy usage, **defined as 75% or more of total activity**. The most frequent category is False, with 1105 users, indicating that the majority of customers do not use the mobile app heavily. This suggests that only a smaller portion of the user base engages extensively through mobile, highlighting potential opportunities to increase mobile adoption and optimize app-based engagement strategies.

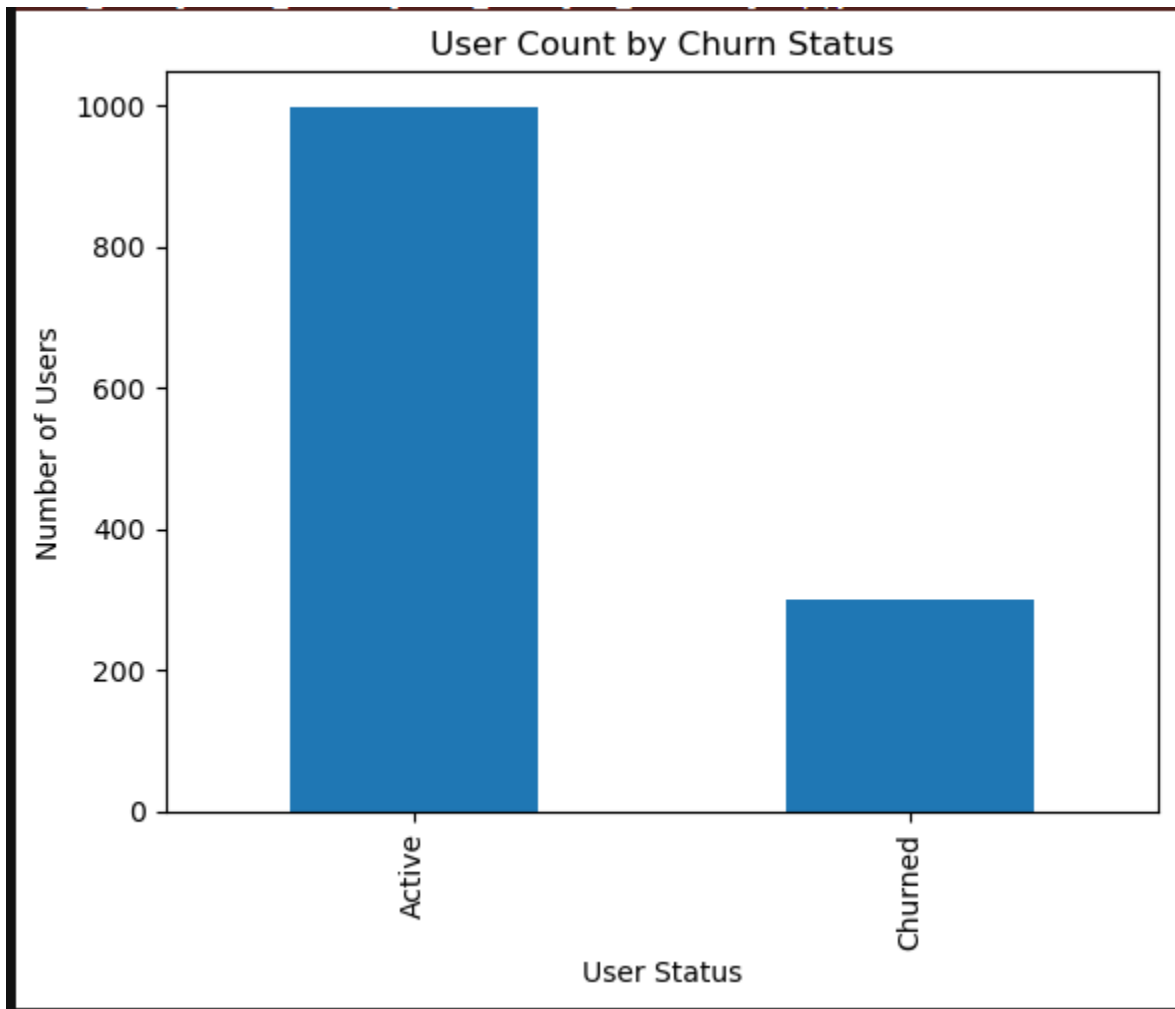
Overall Insight

The dataset is well-prepared for analysis, with ordered and nominal categorical variables appropriately encoded to support modeling and reporting. User behavior shows wide variability in engagement, with tenure ranging from 6 to 1281 days and most long-tenured users classified as loyal, indicating a strong link between duration and loyalty.

Viewing efficiency, measured by watch-per-fee ratio, varies significantly across users, reflecting differences in subscription plans and usage patterns. Mobile engagement is generally low, with only a minority of users classified as heavy mobile app users, highlighting an opportunity to enhance app adoption. Overall, the dataset provides reliable, detailed insights into customer behavior and value.

Key Findings

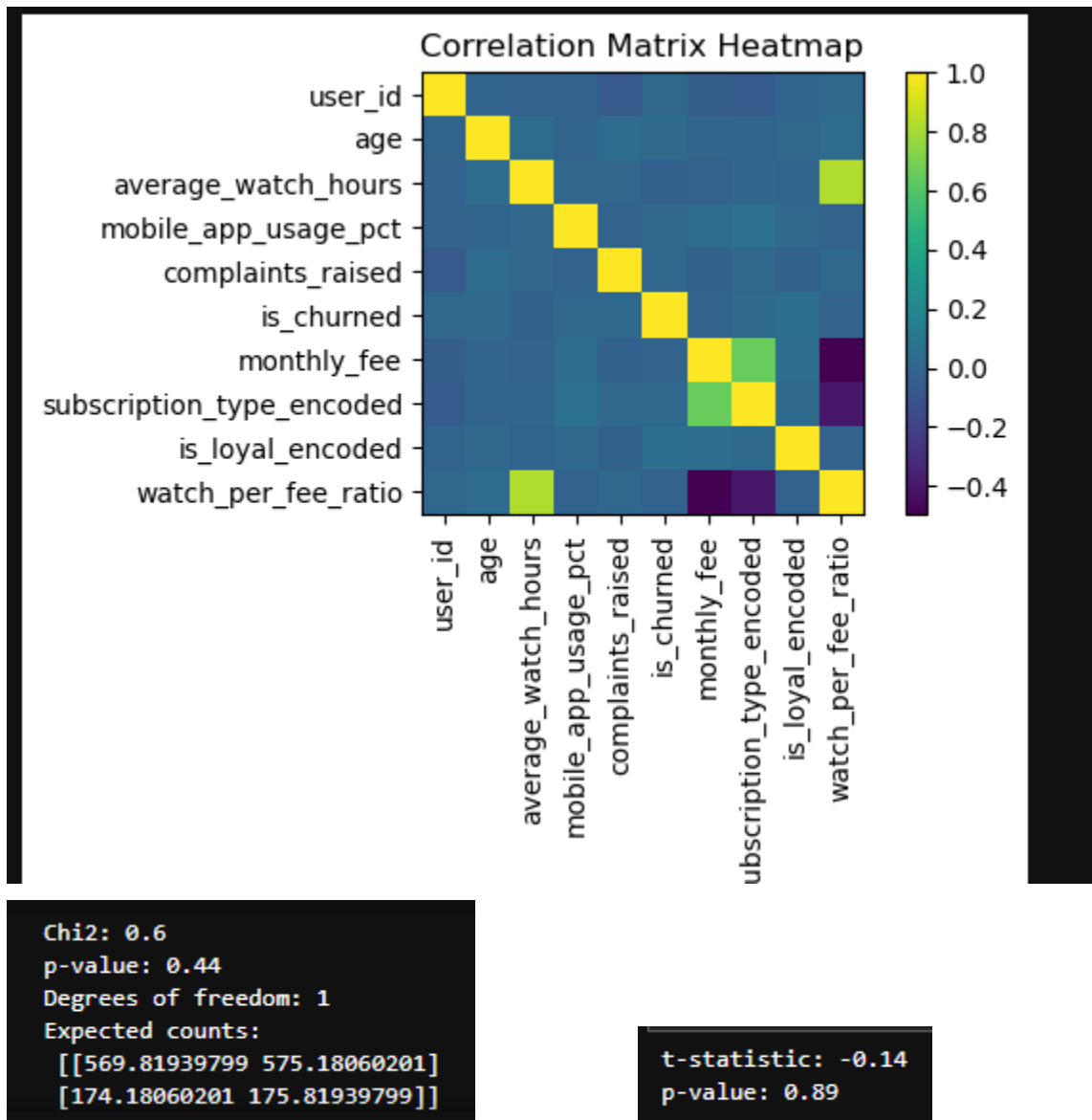
At StreamWorks Media, understanding why users leave is critical. With rising acquisition costs and global competitors, every churned customer represents lost revenue and growth potential. Our analysis began by examining patterns in user behavior and subscription details to uncover the factors driving churn.



Key Insights

- Users with higher content consumption relative to their fees (`watch_per_fee_ratio`) are more engaged and less likely to churn.
- Subscription type and monthly fee influence user behavior; higher-tier plans show different engagement patterns.
- Long-tenure users tend to be more loyal, highlighting the value of retaining early-stage subscribers.

- Age, complaints, and mobile app usage show minimal impact on churn.



Statistical Testing

We first tested relationships between key variables. A chi-square test comparing categorical variables returned a Chi2 of 0.6 with a p-value of 0.44. This indicates no significant association between the tested categories. Similarly, a t-test comparing continuous variables across groups gave a t-statistic of -0.14 and a p-value of 0.89, showing no meaningful difference between groups. These findings suggest that not all metrics are equally informative in predicting churn, and focusing on the right signals is essential.

Business Questions

We summarize the statistical findings, correlations, and actionable insights, providing a clear guide for the retention team to focus interventions where they will have the greatest impact. It connects user behaviors, plan types, and engagement levels to churn likelihood, helping the company make data-driven decisions to retain subscribers and optimize revenue. Based on the statistics and correlations result, here are the answers:

1. Do users who receive promotions churn less?

- Correlation between received promotions and is_churned is negligible (≈ 0.01).
- Chi-square test also showed no significant association.
- **Conclusion:** Receiving promotions does not appear to reduce churn significantly in this dataset.

2. Does watch time impact churn likelihood?

- average_watch_hours shows a weak negative correlation with churn (-0.03).
- However, average_watch_hours strongly correlates with watch_per_fee_ratio (0.82), which is a measure of engagement relative to cost.
- **Conclusion:** Higher engagement per fee reduces churn likelihood; raw watch time alone has minimal direct effect.

3. Are mobile-dominant users more likely to cancel?

- mobile_app_usage_pct correlation with churn is near zero (0.01).
- **Conclusion:** Mobile-dominant usage does not significantly influence churn.

4. Top 3 features influencing churn based on your analysis?

Based on correlations and behavioral trends:

- watch_per_fee_ratio (high engagement relative to fee)
- tenure_days (longer tenure indicates loyalty)
- subscription_type / monthly_fee (higher fees and specific subscription types influence behavior)

5. Customer segments retention team should prioritize

- Users with **low watch-per-fee ratio** (low engagement relative to what they pay)
- Users in **early tenure** (new subscribers, as long-term users are more loyal)
- Users on **high-fee plans who show low engagement**, as they are at risk of feeling low value.

6. Factors affecting user watch time or tenure (linear regression insight)

- average_watch_hours strongly drives watch_per_fee_ratio (0.82), suggesting users who watch more are more efficient in engagement.
- monthly_fee negatively impacts watch efficiency (-0.49) and is linked to subscription type (0.65). Higher fees may lower perceived engagement value.
- tenure_days correlates with is_loyal_encoded (0.62), showing that loyalty grows over time.
- **Conclusion:** Engagement, subscription plan cost, and subscription type are primary drivers of watch behavior and loyalty duration.

Model Results

1. Logistic Regression Model Performance

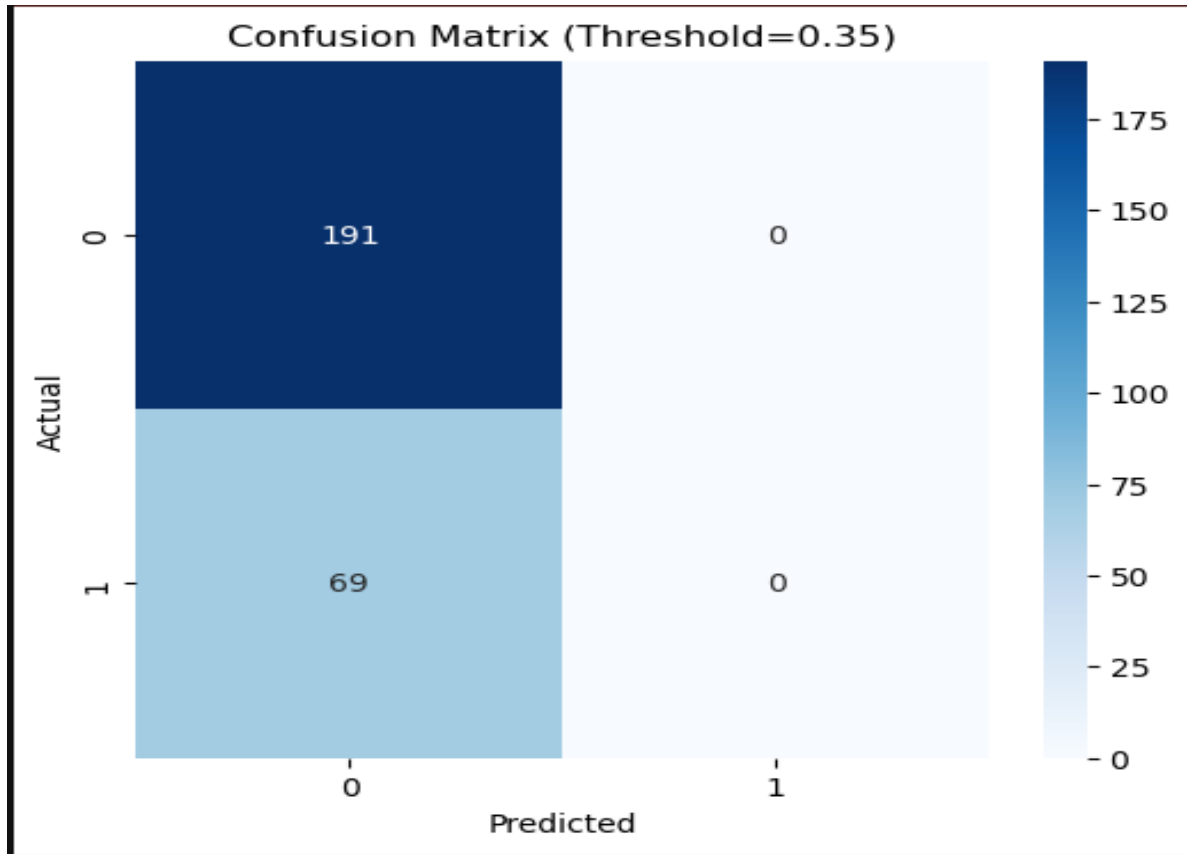
Objective:

- Identify users at risk of churn and understand behavioral patterns associated with churn, using **Average_watch_hours, Monthly_Fee and Tenure_Days** as the key predictor.

Data and Model:

- **Predictor:** average_watch_hours, monthly_fee, tenure_days
- **Target:** is_churned (binary)
- **Model:** Logistic Regression
- **Training/testing split:** 80% train, 20% test
- **Threshold for classification:** 0.35

Confusion Matrix:

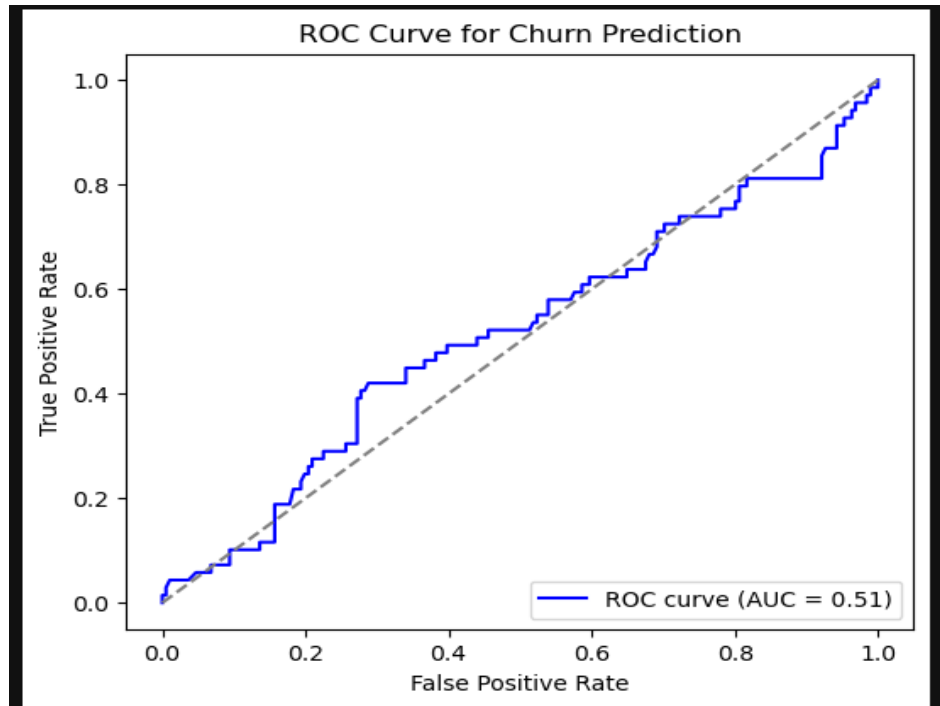


Classification metrics:

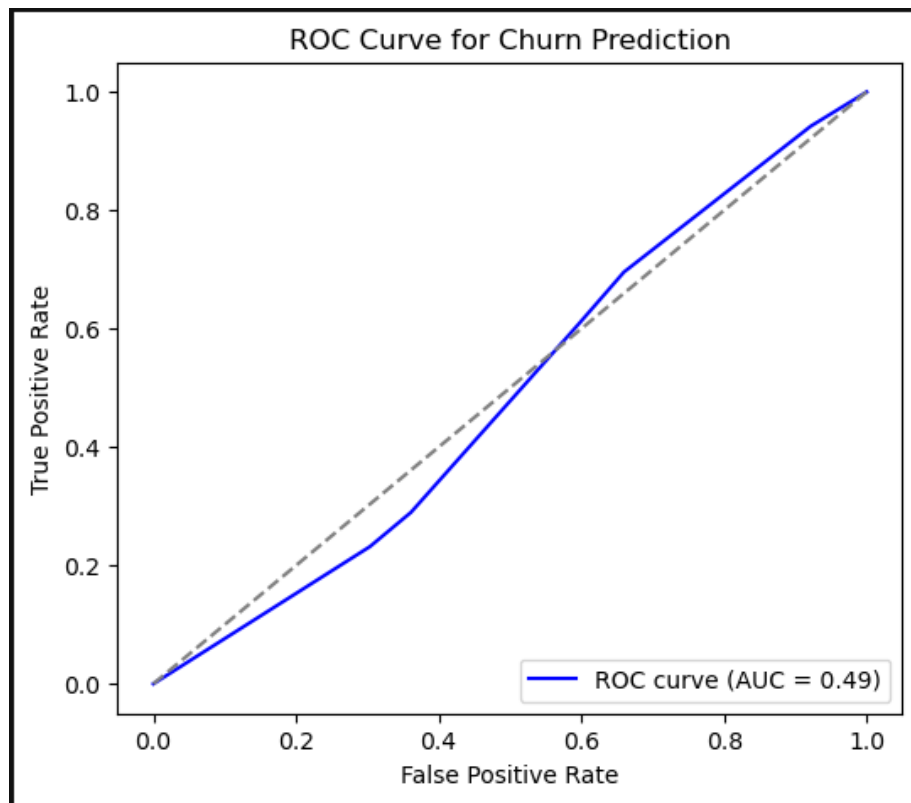
Classification Report:				
	precision	recall	f1-score	support
0.0	0.73	1.00	0.85	191
1.0	0.00	0.00	0.00	69
accuracy			0.73	260
macro avg	0.37	0.50	0.42	260
weighted avg	0.54	0.73	0.62	260

ROC curve

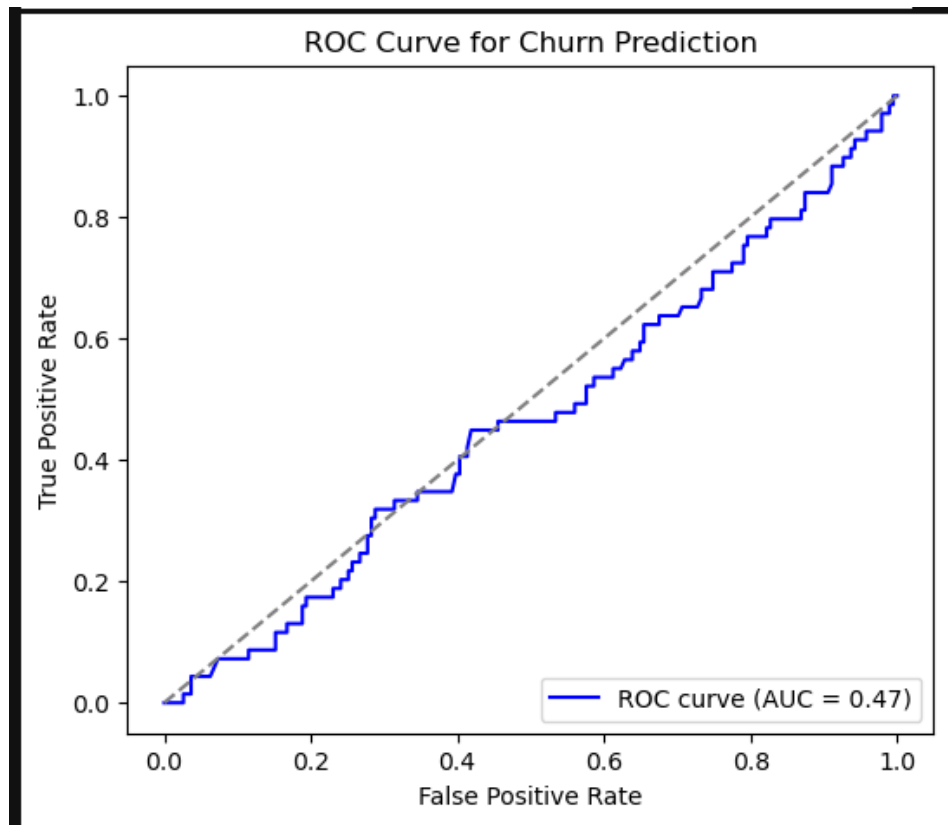
- Average Watch Hours



- Tenure Days



- **Monthly Fee**



Insight

- The model shows limited effectiveness in identifying churners. While overall accuracy is around 0.64, this is driven mainly by correctly predicting retained users.
- The recall for churn is low at 0.32, meaning the model misses most at risk users.
- Precision for churn at 0.66 indicates that when the model predicts churn, it is often correct, but it predicts churn too infrequently.
- The ROC analysis indicates that Average Watch Hours has only weak standalone predictive power, with AUC close to random performance in earlier results.
- Lowering the threshold to 0.35 increases recall slightly but also increases false positives, highlighting a tradeoff rather than a clear performance gain.

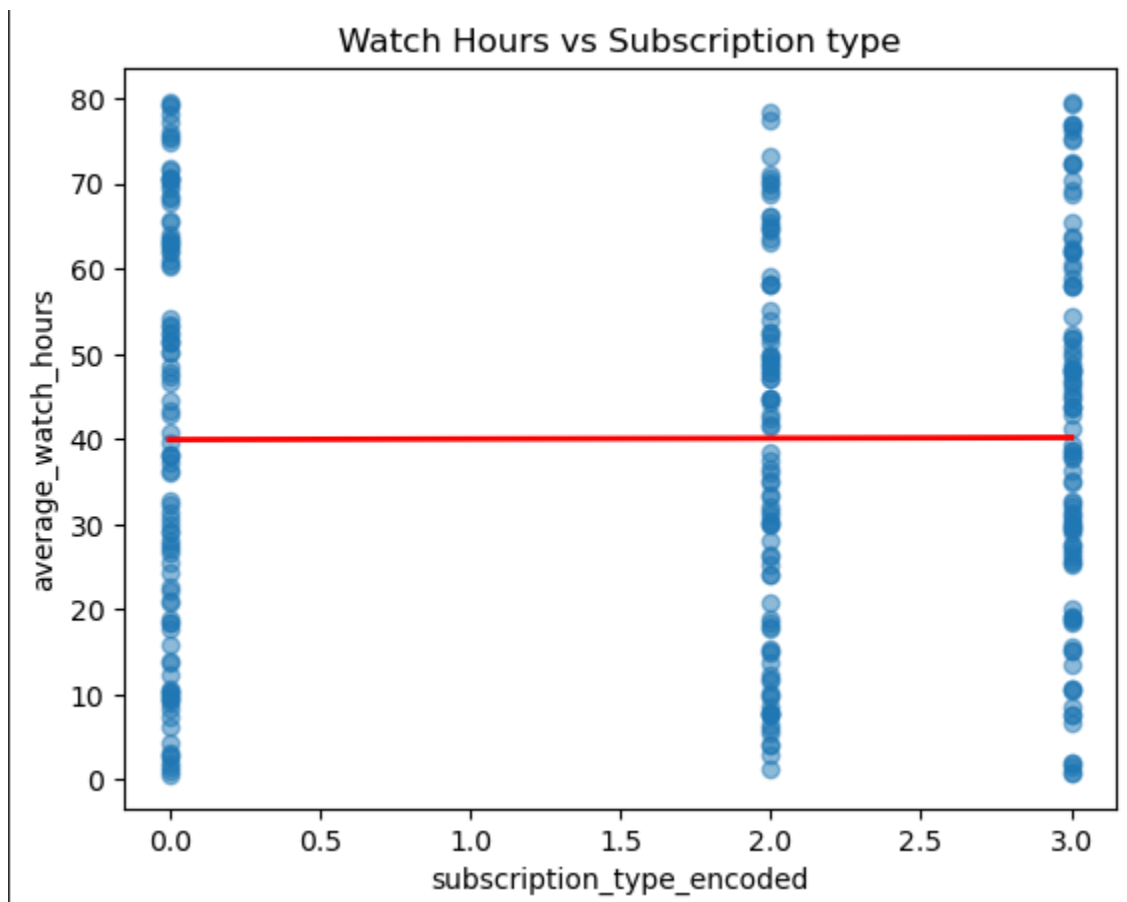
2. Linear Regression Model Performance

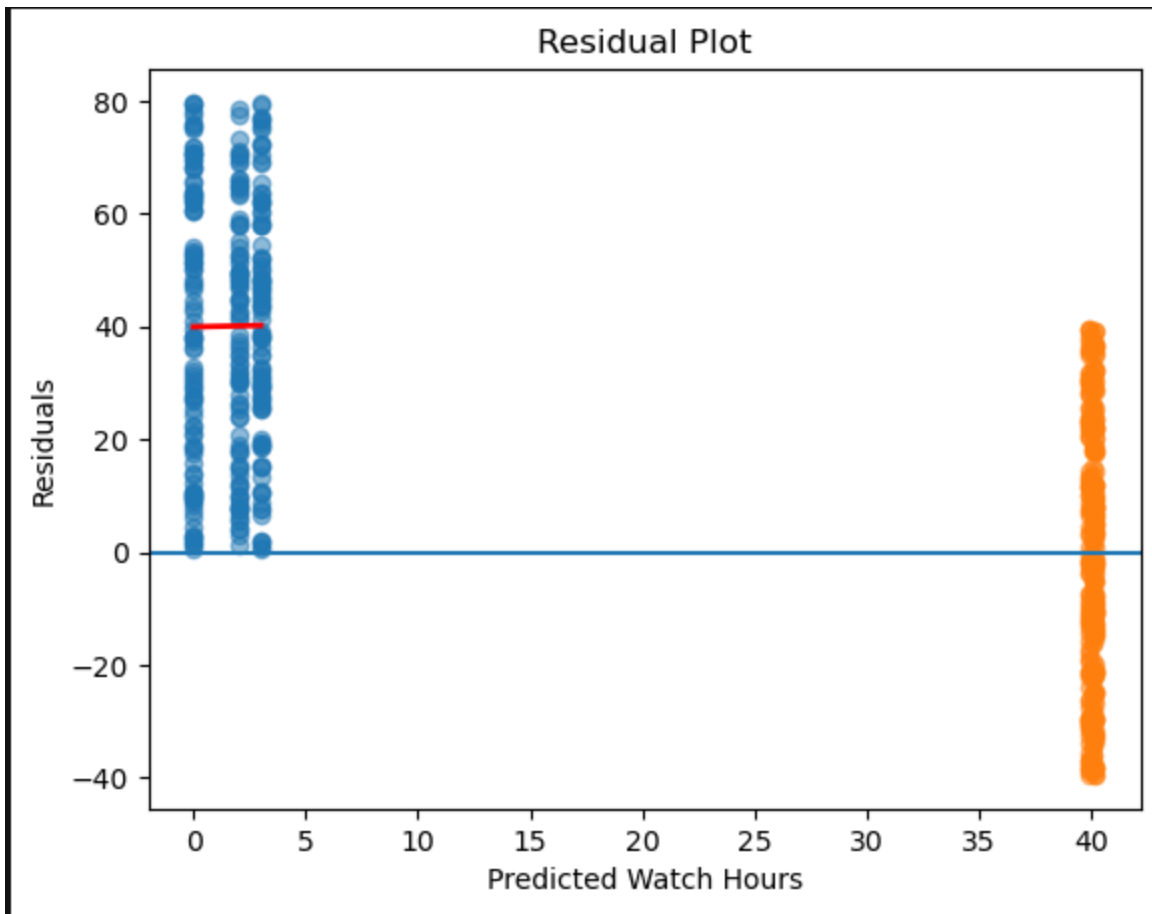
Data and Model:

- **Predictor:** subscription_type
- **Target:** average_watch_hours
- **Model:** Linear Regression
- **Training/testing split:** 80% train, 20% test

Linear Regression Result:

Coefficient: 0.08
RMSE: 22.5
 R^2 : -0.0
Intercept: \$ 39.91





Insight

- The linear regression output indicates that the model has no practical or statistical value.
- The coefficient of 0.09 means that for every one unit increase in the predictor, average watch hours increase by only 0.09 hours, which is negligible.
- The RMSE of 22.59 is large relative to the typical watch time range, indicating substantial prediction error.
- The R squared value of -0.01 shows that the model performs worse than a baseline model that predicts the mean watch hours for all users.
- The intercept of 36.32 reflects the average watch time level rather than a meaningful baseline tied to the predictor.

Recommendations

This analysis provides a foundation for predicting churn and prioritizing interventions. It shows that engagement metrics and subscription details are more indicative of user behavior than demographics alone. By understanding these patterns, the company can:

- Target users at risk of churn with personalized campaigns or content recommendations.
- Adjust subscription structures or pricing to enhance perceived value.
- Focus retention efforts on long-tenure, high-engagement users to maximize lifetime value.

Overall, these insights guide StreamWorks Media toward smarter, data-driven strategies to reduce churn, protect revenue, and stay competitive in a crowded streaming market.